3. Chemical adsorption is fast with increase pressure, it can not alter the amount.	3. In Physisorption, when pressure increases the extent of adsorption increases.
4. When temperature is raised chemisorption first increases and then decreases.	4. Physisorption decreases with increase in temperature.
5. Chemisorption involves transfer of electrons between the adsorbent and adsorbate.	5. No transfer of electrons
6. Heat of adsorption is high i.e., from 40-400kJ/mole.	6. Heat of adsorption is low in the order of 40kJ/mole.
7. Monolayer of the adsorbate is formed.	7. Multilayer of the adsorbate is formed on the adsorbent.
8. Adsorption occurs at fixed sites called active centres. It depends on surface area	8. It occurs on all sides.
9. Chemisorption involves the formation of activated complex with appreciable activation energy.	9. Activation energy is insignificant.

10.1.2 Factors affecting adsorption

The adsorption is well understood by considering the various factors affecting it.

Qualitatively, the extent of surface adsorption depends on the following factors

(i) Nature of adsorbent

(ii) Nature of adsorbate

(iii) Pressure

(iv) Concentration at a given temperature.

1. Surface area of adsorbent:

As the adsorption is a surface phenomenon it depends on the surface area of adsorbent. i.e., higher the surface area, higher is the amount adsorbed.

2. Nature of adsorbate

The nature of adsorbate can influence the adsorption. Gases like SO_2,NH_3,HCl and CO_2 are easily liquefiable as they have greater van der waal's force of attraction. On the other hand, permanent gases like H_2,N_2 and O_2 can not be liquefied easily. These permanent gases are having low critical temperature and adsorbed slowly, while gases with high critical temperature are adsorbed readily.

XII U10-Surface Chemistry.indd 72 11-01-2021 15:33:26